Notrixum Injection 10mg/ml
Atracurium Besylate

Product Description
Clear and colourless solution for injection

Composition
Each ml of injection contains Atracurium Besylate 10 mg.

Pharmacodynamics
Atracurium is a competitive non-depolarising neuromuscular-blocking agent which acts by competing with acetylcholine for occupancy of receptors on the post-synaptic membrane. Thus, the receptors sites are blocked by Atracurium resulting in failure of contraction and eventual flaccid paralysis.

Pharmacokinetics
The pharmacokinetics of Atracurium in man are essentially linear with the 0.3-0.6 mg/kg dose range. The elimination half-life is approximately 20 minutes, and the volume of distribution is 0.16 L/kg. Atracurium is 82% bound to plasma proteins.

Atracurium is degraded spontaneously mainly by a non-enzymatic decomposition process (Hofmann elimination) which occurs at plasma pH and at body temperature and produces breakdown products which are inactive. Degradation also occurs by ester hydrolysis catalysed by non-specific esterases. Elimination of atracurium is not dependent on kidney or liver function.

The main breakdown products are laudanosine and a monoquaternary alcohol which have no neuromuscular blocking activity. The monoquaternary alcohol is degraded spontaneously by Hofmann elimination and excreted by the kidney. Laudanosine is excreted by the kidney and metabolised by the liver. The half-life of laudanosine ranges from 3-6 hours in patients with normal kidney and liver function. It is about 15 hours in renal failure and is about 40 hours in renal and hepatic failure. Peak plasma levels of laudanosine are highest in patients without kidney or liver function and average 4 µg/ml with wide variation.

Concentration of metabolites are higher in ICU patients with abnormal renal and/or hepatic function. These metabolites do not contribute to neuromuscular block.

Indications
Atracurium Besylate is a highly selective, competitive or non-depolarizing neuromuscular-blocking agent which is used as an adjunct to general anaesthesia to enable tracheal intubations to be performed and to relax skeletal muscles during surgery or controlled ventilation; to facilitate mechanical ventilation in Intensive Care Unit (ICU) patients.

Recommended Dosage
Use by Injection in Adults: Notrixum is administered by IV Injection/IV Infusion.

Dosage Range for Adults:

0.3 – 0.6 mg/kg (depending on the duration of full block required) and will provide relaxation for about 15 – 35 minutes. Endotracheal intubation can usually be accomplished within 90 seconds from I.V injection of 0.5 – 0.6 mg/kg.

Full block can be prolonged with supplementary doses of 0.1 – 0.2 mg/kg as required. Successive supplementary dosing does not give rise to accumulation of neuromuscular-blocking effect. Spontaneous recovery from the end of full block occurs in about 35 minutes as measured by the restoration of the tetanic response to 95 % of normal neuromuscular function. The neuromuscular block produced by Notrixum can be rapidly reversed by standard doses of anticholinesterase agents, e.g neostigmine and edrophonium, accompanied or preceded by atropine, with no evidence of re-curarisation.

Use as an Infusion in Adults:
After an initial bolus dose of 0.3 – 0.6 mg/kg, Notrixum can be used to maintain neuromuscular block during long surgical procedures by administration as a continuous infusion at rates of 0.3 – 0.6 mg/kg/hr.

Notrixum can be administered by infusion during cardiopulmonary bypass surgery at the recommended infusion rates. Induced hypothermia to a body temperature of 25 – 26°C reduces the rate of inactivation of Atracurium Besylate, therefore, full neuromuscular block may be maintained by approximately half the original infusion rate at these low temperatures.

Notrixum is compatible with the following infusion solutions for the times stated below:

Infusion Solution	Period of Stability
NaCl infusion 0.9% w/v	24 hours
Glucose infusion 5.0% w/v	8 hours
Ringer’s Injection	8 hours
NaCl 0.18% w/v & glucose 4.0% w/v	8 hours
Hartmann’s solution for injection	4 hours

When diluted in these solutions, the resultant solutions will be stable in daylight for the stated periods at room temperatures.

Children
The dosage in children > 1 month is the same as that in adult on a body weight basis.

Elderly
Notrixum may be used at standard dosage in elderly patients. It is recommended, however, that the initial dose be at the lower end of the range and that it be administered slowly.

Patients with Reduced Renal and/or Hepatic Function:
Notrixum may be used at standard dosage at all levels of renal or hepatic function, including end-stage failure.

Patients with Cardiovascular Disease:
In patients with clinically significant cardiovascular disease, the initial dose of Notrixum should be administered over a period of 60 seconds.

Intensive Care Unit (ICU) Patients:
After an optional initial bolus dose of 0.3 – 0.6 mg/kg, Notrixum can be used to maintain neuromuscular block by administering a continuous infusion at rates of between 11 and 13 μg/kg/minute (0.65 – 0.78 mg/kg/hour). However, there is wide interpatient variability in dosage requirements.

Dosage requirements may change with the time. Infusion rates as low as 4.5 μg/kg/minute (0.27 mg/kg/hour) or as high as 29.5 μ g/kg/minute (1.77 mg/kg/hour) are required in some patients. The rate of spontaneous recovery from neuromuscular block after infusion of Notrixum in ICU Patients is independent of the duration of administration. Spontaneous recovery to a train-of-four-ratio > 0.75 (the ratio of the height of the 4th to the 1st twitch in a train-of-four) can be expected to occur in approximately 60 minutes. A range of 32 – 108 minutes has been observed in clinical trials.

Monitoring:
In common with all neuromuscular-blocking agents, monitoring of neuromuscular function is recommended during the use of Notrixum in order to individual dosage requirements.

Incompatibilities
None

Route of Administration
Intravenous

Contraindications
Patients known to have an allergic hypersensitivity to Notrixum.

Warnings and Precautions
In common with all other neuromuscular-blocking agents, Notrixum paralyses the respiratory muscles as well as other skeletal muscles, but has no effect on consciousness. Notrixum should be administered only with adequate general anaesthesia and only by or under the close supervision of an experienced anaesthetist with adequate facilities for endotracheal intubation and artificial ventilation. In common with other neuromuscular-blocking agents, the potential for histamine release exists in susceptible patients during Notrixum administration. Caution should be exercised in administering Notrixum in patients with a history suggestive of an increased sensitivity to the effect of histamine. Notrixum does not have significant vagal or ganglionic-blocking properties in the recommended dosage range. Consequently, Notrixum has no clinically significantly effects on heart rate in the recommended dosage range and it will not counteract the bradycardia produced by many anaesthetic agents or by vagal stimulation during surgery.

In common with other non-depolarising neuromuscular-blocking agents, increased sensitivity to Atracurium may be expected in patients with myasthenia gravis, other forms of neuromuscular disease and severe electrolyte imbalance.

Notrixum should be administered over a period of 60 seconds to patients who may be unusually sensitive to fall in arterial blood pressure, e.g. those who are hypovolemic.

When a small vein is selected as the injection side, Atracurium should be flushed through the vein with physiological saline after injection. When other anaesthetic drugs are administered through the same in dwelling needle or cannula as Notrixum is important that each drug is flushed through with an adequate volume of physiological saline.

Atracurium Besylate is hypotonic and must not be administered into the infusion line of a blood transfusion.

Studies of malignant hyperthermia in susceptible animals (swine) and clinical studies in patients susceptible to malignant hyperthermia indicate that Notrixum, does not trigger this syndrome.

In common with other non-depolarising neuromuscular-blocking agents, resistance may develop in patients suffering from burns. Such patients may require increased doses dependent on the time elapsed since the burn injury and the extent of the burn.

Carcinogenicity, Mutagenicity and Impairment of Fertility:
Atracurium has been evaluated in 3 short term mutagenicity test. It was not mutagenic in either the in vitro Ames salmonella assay at concentrations up to 1000 µg/plate or in an in vivo rat bone marrow assay at doses up to those which resulted in neuromuscular blockade. In a 2nd in vitro test, the mouse lymphoma assay, mutagenicity was not observed at doses up to 60 µg/ml which killed up to 50 % of the treated cell, but it was moderately mutagenic at concentrations of 80 µg/ml in the absence of metabolizing agent and weakly mutagenic at very high concentrations (1200 µg/ml) when metabolizing enzymes were added. At both concentrations, > 80 % of the cells were killed.

In view of the nature of human exposure to Atracurium, the mutagenic risk to patients undergoing surgical relaxation with Notrixum must be considered negligible. Fertility studies have not been performed.

Cautions for Usage
Notrixum is inactivated by high pH and so must not be mixed in the same syringe with thiopentone or any alkaline agent. When a small vein is selected as the injection site, Notrixum should be flushed through the vein with physiological saline after injection. When other anaesthetic drugs are administered through the same in-dwelling needle or cannula as Notrixum, it is important that each drug is flushed through with an adequate volume of physiological saline.

Interactions with other Medicaments

- The neuromuscular block produced by Notrixum may be increased by the concomitant use of inhalation anaesthetics, e.g. halothane, isoflurane and enflurane.
- In common with all non-depolarising neuromuscular-blocking agents, the magnitude and/or duration of non-depolarizing neuromuscular block may be increased as a result of interaction with :

1. Antibiotic: Aminoglycosides, Polymyxins, Spectinomycin, Tetracyclines, Lincomysin and Clindamycin.
2. Antiarrhythmic Drugs: Propranolol, Calcium-Channel Blockers, Lignocaine, Procainamide and Quinidine.
3. Diuretic: Furosemide and possibly mannitol, thiazide diuretics and acetazolamide.
4. Magnesium Sulphate
5. Ketamine
6. Lithium Salts
7. Ganglion-Blocking Agents: Trimetaphan, Hexamethonium.

Rarely, certain drugs may aggravate or unmask latent myasthenia gravis or actually induce a myasthenic syndrome; increased sensitivity to Notrixum would be consequent on such a development. Such drugs include various antibiotics, β - blockers (propranolol, oxprenolol), antiarrhythmic drugs (procainamide, quinidine), antirheumatic drugs (chloroquine, D – penicillamine), trimetaphan, chlorpromazine, steroids, phenytoin and lithium. The onset of non-depolarizing neuromuscular block is likely to be lengthened and the duration of block shortened in patients receiving chronic anticonvulsant therapy. The administration of combinations non-depolarising neuromuscular blocking agents in excess of that which might be expected were an equipotent total dose of Notrixum was administered. Any synergistic effect may vary between different drug combinations. A depolarizing muscle relaxant, e.g. suxamethonium chloride should not be administered to prolong the neuromuscular blocking effects of non-depolarising agents, e.g. Atracurium, as this may result in a prolonged and complex block which can be difficult to reverse with anticholinesterase drugs.

Pregnancy and Lactation
In common with all neuromuscular-blocking agents, Notrixum should be used during pregnancy only if the potential benefit to the mother outweighs any potential risk to the foetus.

Notrixum is suitable for maintenance of muscle relaxation during caesarean section as it does not cross the placenta in clinically significant amounts following recommended doses.

It is not known whether Notrixum is excreted in human milk.

Atracurium Besylate is inactivated by high pH and so must not be mixed in the same syringe with Thiopentone or any alkaline agent.

Side Effects
Associated with the use of Notrixum, there have been reports of skin flushing, mild transient hypotension or bronchospasm, which have been attributed to histamine release. Very rarely, severe anaphylactoid reactions have been reported in patients receiving Notrixum in conjunction with one or more anaesthetic agents.

There have been some reports of seizures in ICU patients who have been receiving Atracurium concurrently with several other agents. These patients usually had one or more medical conditions predisposing to seizures (e.g. cranial trauma, cerebral oedema, viral encephalitis, hypoxic encephalopathy, and uraemia). In clinical trials, there appears to be no correlation between plasma laudanosine concentration and the occurrence of seizures.

There have been some reports of muscle weakness and/or myopathy following prolonged use of muscle relaxants in severely ill patients in the ICU. Most patients were receiving concomitant corticosteroids. These events have been seen infrequently in association with Notrixum. A causal relationship has not been established.

Overdose and Treatment
Symptoms: Prolonged muscle paralysis and its consequences are the main signs of over dosage.

Treatments: It is essential to maintain a patent airway together with assisted positive pressure ventilation until spontaneous respiration is adequate. Full sedation will be required since consciousness is not impaired. Recovery may be hastened by the administration of anticholinesterase agents accompanied by atropine or glycopyrrolate, once evidence of spontaneous recovery is present.

Storage Conditions
Store at cold room (2 - 8ºC) to preserve potency. Do not freeze.
Upon removal from refrigerator to room temperature storage conditions (below 30ºC), use Notrixum injection within 14 days even if refrigerated.

Dosage Forms and Packaging Available
NOTRIXUM® 25 INJECTION: Box contain of 5 ampoules @ 2.5 ml.
NOTRIXUM® 50 INJECTION: Box contain of 5 ampoules @ 5 ml.

Shelf-life
24 months

Registration Number
Notrixum® Injection 10 mg/ml – MAL15045128AZ

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